Running the primary experiment

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

WORKBOOK · 1 OF 15

Recall and choose your starting point

> Record your starting explanation for Running the primary experiment before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Reproducible environment and records

Without opening the old lesson, explain one key idea from Reproducible environment and records. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Pilot experiments

Without opening the old lesson, explain one key idea from Pilot experiments. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Research ethics and consent

Without opening the old lesson, explain one key idea from Research ethics and consent. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 15

Practise with fading help

> Complete the supported Running the primary experiment trace and record the first step you can explain without help.

Use W2: Trace the worked case for Running the primary experiment. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 15

Individual reasoning check

> Check the proposed Running the primary experiment explanation against the supplied facts before discussing it.

Use the worked case for Running the primary experiment. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 15

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Apply a new case independently

Twelve cases:9 pass, 1 fail, 2 unassessed. Report same metrics.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 15

W1 Explain the mechanism

> Explain the mechanism. Apply the supplied Running the primary experiment facts and keep your reasoning.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **My reasoning and evidence**

WORKBOOK · 6 OF 15

W2 Trace the worked case

> Trace the worked case. Apply the supplied Running the primary experiment facts and keep your reasoning.

Twenty eligible cases:16 pass, 2 fail, 2 unassessed after outage. Plan reports completed rate plus coverage. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

WORKBOOK · 7 OF 15

W3 Audit the limit

> Audit the limit. Apply the supplied Running the primary experiment facts and keep your reasoning.

A report omits this qualification: Following the plan and reporting deviations are required even when results contradict the hypothesis.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

WORKBOOK · 8 OF 15

W4 Apply a new case independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply a new case independently. Apply the supplied Running the primary experiment facts and keep your reasoning.

Twelve cases:9 pass, 1 fail, 2 unassessed. Report same metrics.

My reasoning and evidence

WORKBOOK · 9 OF 15

W5 Design a revealing follow-up

> Design a revealing follow-up. Apply the supplied Running the primary experiment facts and keep your reasoning.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

WORKBOOK · 10 OF 15

Project checkpoint and reflection

> Connect Running the primary experiment to Reproducible AI Research Study, identify a limit and choose the next useful test.

Add a labelled running the primary experiment **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of running the primary experiment.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

WORKBOOK · 11 OF 15

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 12 OF 15

Pictorial case practice: Running the primary experiment

> Run the frozen experiment. A research team freezes versions A and B, labels and scoring before a six-case primary run. Explain the labelled step before checking the changed condition.

Run the frozen experiment. A research team freezes versions A and B, labels and scoring before a six-case **primary run**.

A scores [1,0,1,1,0,1]; B scores [1,1,1,0,1,1]. Both answer the same six cases. Binary entries mean 1=correct and 0=incorrect.

AI ACADEMY / AI-7/25

Paired primary case outcomes

	Case 1	Case 2	Case 3	Case 4	Case 5	Case 6
Version A	1	0	1	1	0	1
Version B	1	1	1	0	1	1

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/25

Running the primary experiment

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Keep versions fixed

No case result may trigger tuning during this run.

2. Compute results

A totals 4/6; B totals 5/6.

3. Keep case-level records

Cases two and five favour B; case four favours A.

CHANGE ONE THING

Tune B after inspecting its case-four failure, then rescore the same set.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Compute A's and B's six-case counts and identify the cases that favour each. How must a B revision tuned on case four be labelled when evaluated again on the same set?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning.

Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Reproducible AI Research Study record.

Project: Reproducible AI Research Study

Current milestone: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

WORKBOOK · 14 OF 15

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Run the frozen experiment. A research team freezes versions A and B, labels and scoring before a six-case **primary run**.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Cases two and five favour B; case four favours A.

1. Compute results

2. Keep versions fixed

3. Keep case-level records

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Keep case-level records Cases two and five favour B; case four favours A.

Running the primary experiment means following the declared protocol and retaining raw outcomes. Overall scores alone hide different failure patterns.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Tune B after inspecting its case-four failure, then rescore the same set. Which conclusion is supported by the supplied facts?

1. Label the rerun as development on exposed cases, not untouched primary evaluation.

2. This one example guarantees correct results for every future **input**.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Label the rerun as development on exposed cases, not untouched primary evaluation. The evaluation outcomes have now influenced the version.

Repair a missing building block

Review Reproducible environment and records (ai-7/24). Goal: Explain the **mechanism** and terminology of reproducible environment and records. Short practice: Trace the recorded mean for 2,4,6 and its changed result for 2,4,9.

Which artefacts must accompany the **output** so another team can identify the exact run?. Return to this week and explain what you changed.

Review Pilot experiments (ai-7/22). Goal: Explain the **mechanism** and terminology of pilot experiments. Short practice: Calculate reviewer agreement on the four pilot answers.

What can the disagreement improve, and why are those revised practice answers not untouched final **evidence**?. Return to this week and explain what you changed.

Review Research ethics and consent (ai-7/19). Goal: Explain the **mechanism** and terminology of research ethics and consent. Short practice: Identify which uses the drawing consent permits and excludes.

Can the unrelated project reuse the drawings under this agreement, and what new authorisation would be needed?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

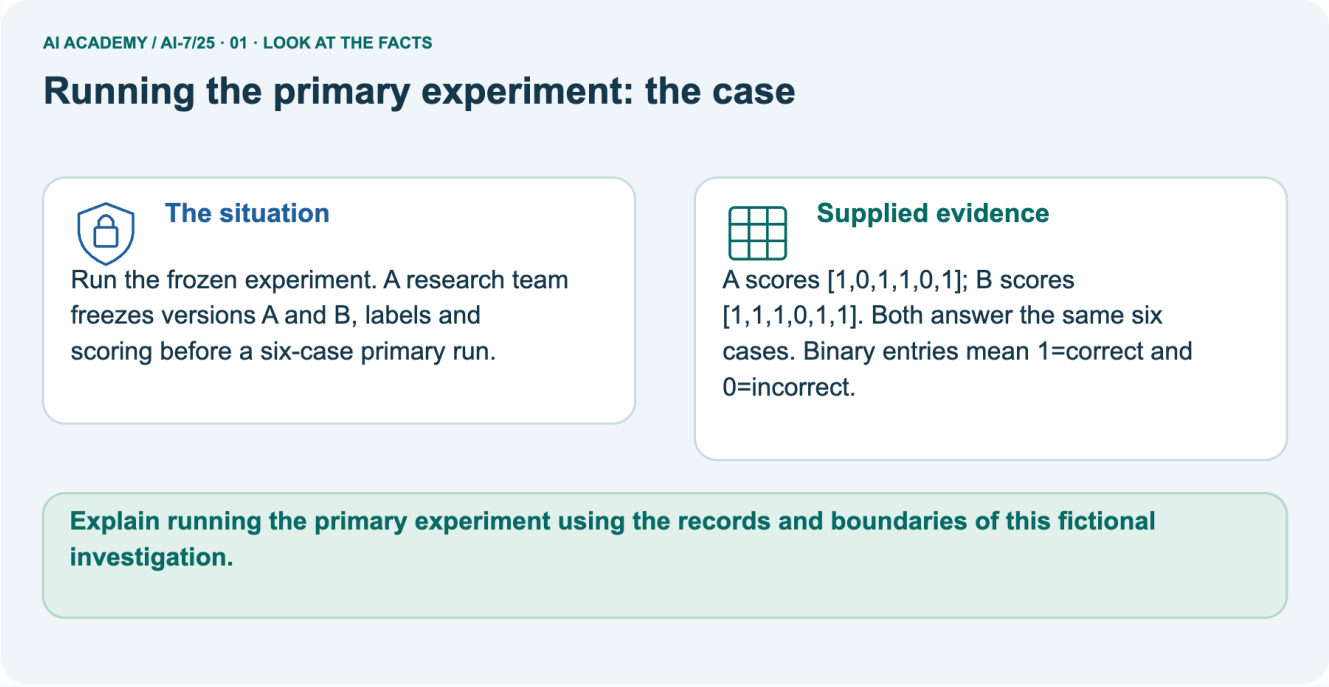
What I repaired, the evidence I kept and the next test

WORKBOOK · 15 OF 15

Visual concept journey

> Explain each Running the primary experiment picture from its labels, then record a relevant limit.

01. Running the primary experiment: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

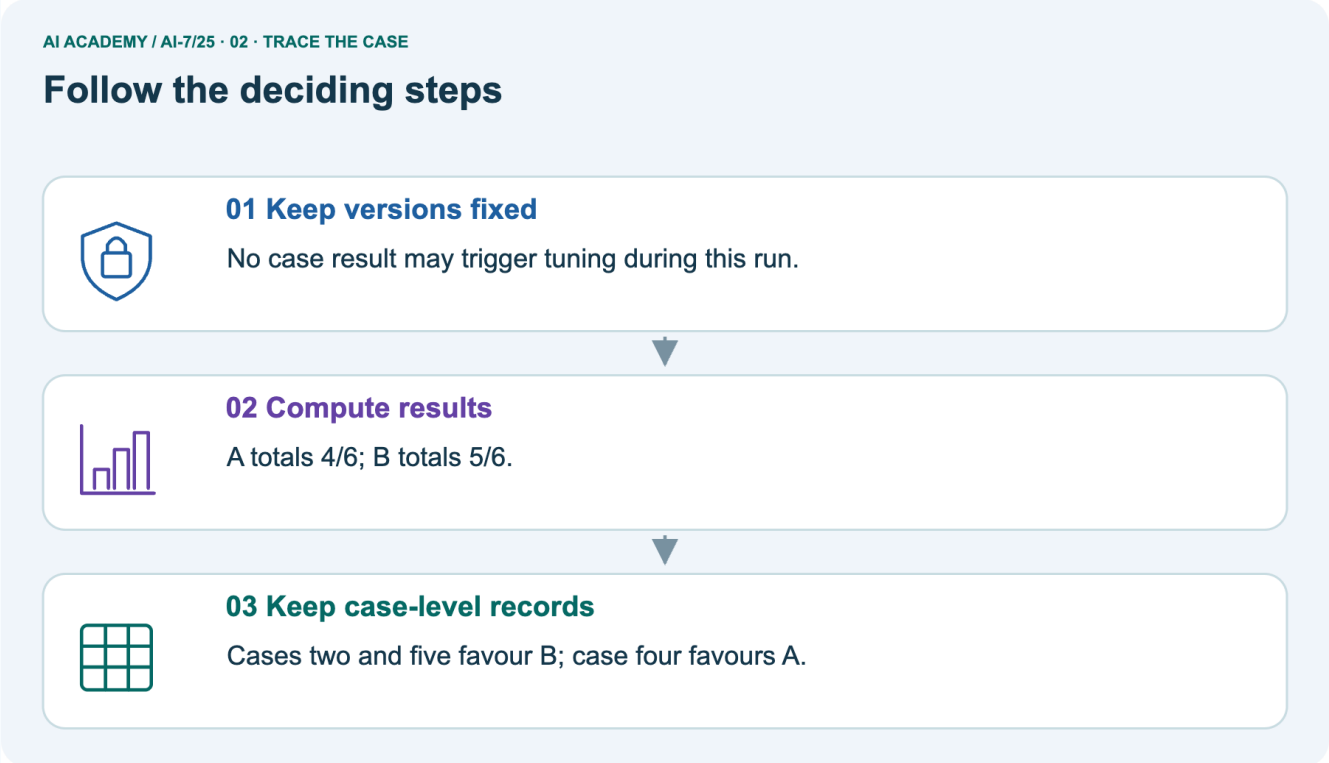
Run the frozen experiment. A research team freezes versions A and B, labels and scoring before a six-case **primary run**. Supplied facts: A scores [1,0,1,1,0,1]; B scores [1,1,1,0,1,1]. Both answer the same six cases.

Binary entries mean 1=correct and 0=incorrect.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Keep versions fixed: No case result may trigger tuning during this run. → Compute results: A totals 4/6; B totals 5/6. → Keep case-level records: Cases two and five favour B; case four favours A.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Paired primary case outcomes



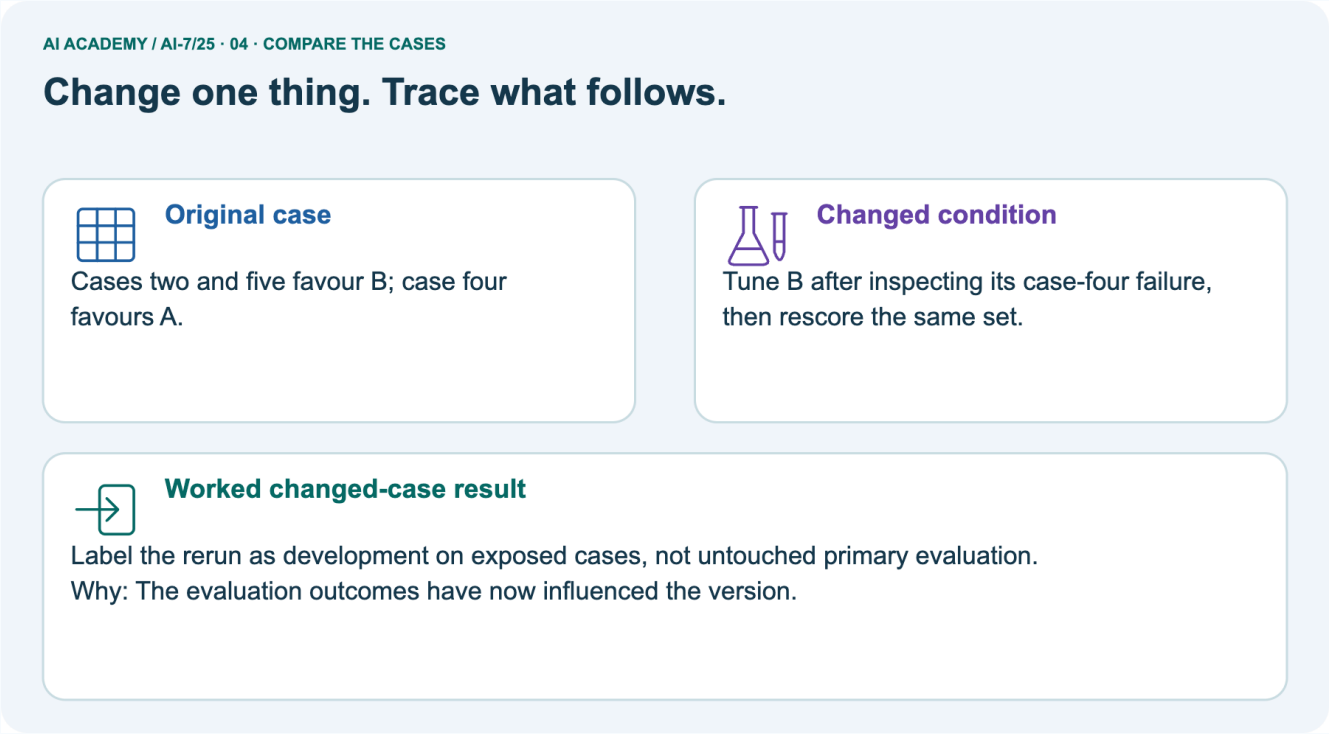
Follow the labelled picture with the matching explanation. Open the vector diagram

Paired primary case outcomes. Version A: Case 1 = 1, Case 2 = 0, Case 3 = 1, Case 4 = 1, Case 5 = 0, Case 6 = 1. Version B: Case 1 = 1, Case 2 = 1, Case 3 = 1, Case 4 = 0, Case 5 = 1, Case 6 = 1.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

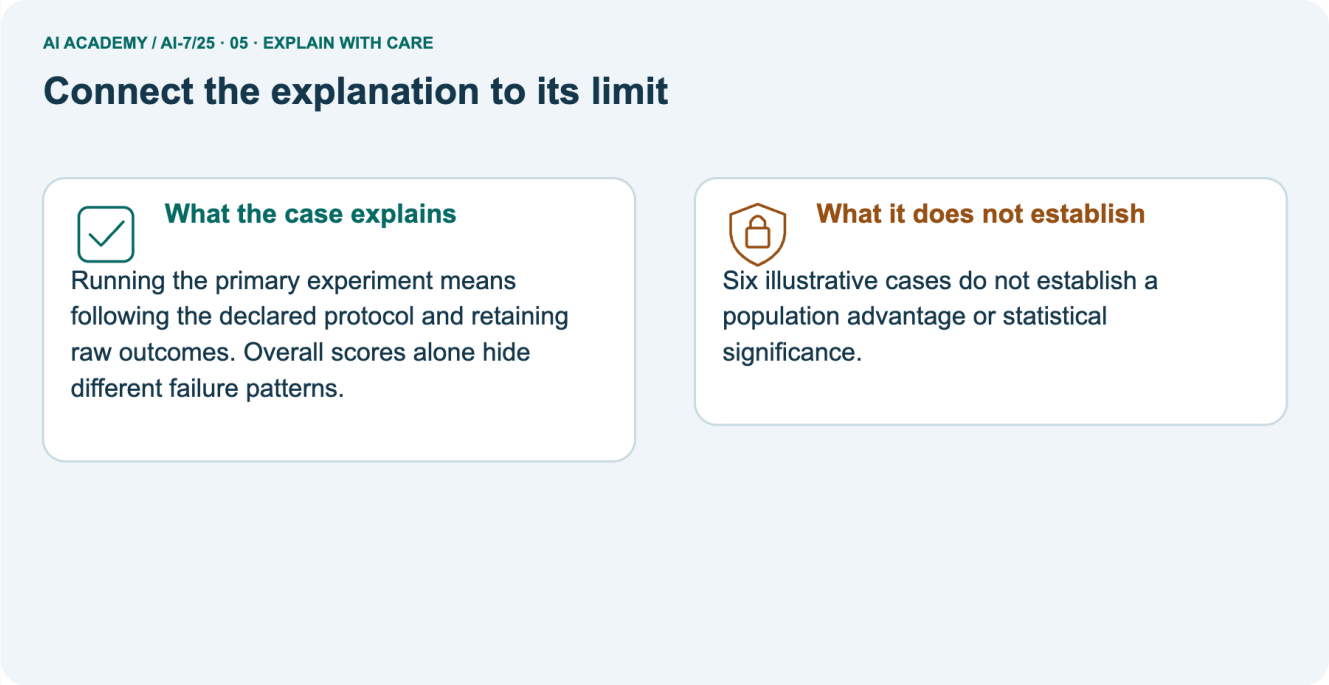
Original case: Cases two and five favour B; case four favours A. Changed condition: Tune B after inspecting its case-four failure, then rescore the same set.

Expected result: Label the rerun as development on exposed cases, not untouched primary evaluation. Reason: The evaluation outcomes have now influenced the version.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

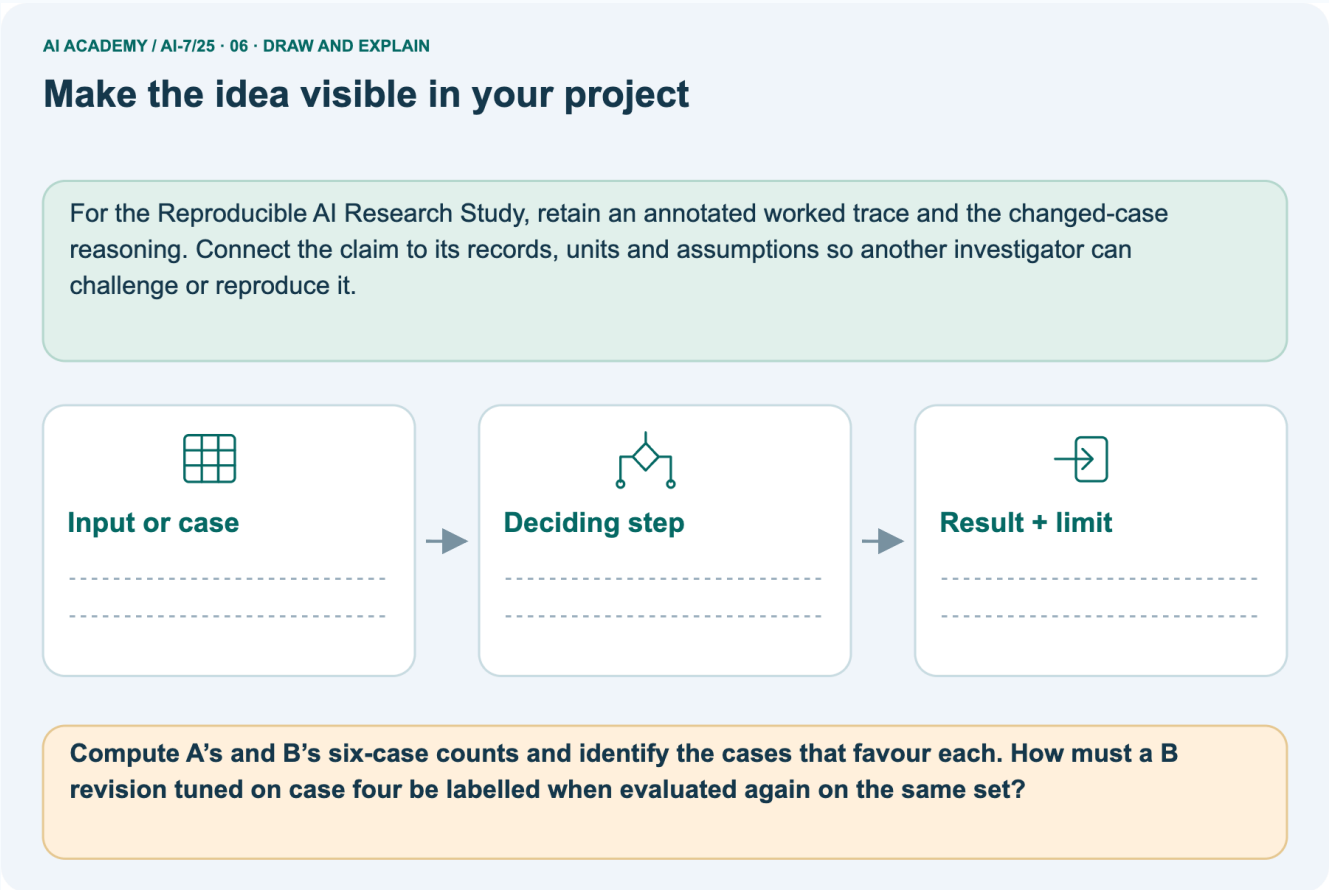
Running the primary experiment means following the declared protocol and retaining raw outcomes. Overall scores alone hide different failure patterns.

Limit: Six illustrative cases do not establish a population advantage or statistical significance.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Compute A's and B's six-case counts and identify the cases that favour each.

How must a B revision tuned on case four be labelled when evaluated again on the same set?

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.